Who Moves First?

Information diffusion between Nasdaq-100 and S&P 500 futures and ETFs, 2016–2026

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The question. When news arrives (an inflation print, a Fed decision, a large order), it does not reach every price at the same instant. The Nasdaq-100 and the S&P 500 each trade as a futures contract (NQ, ES) and as an ETF (QQQ, SPY), all carrying the same information. Which market moves first, has that changed over ten years, what happens around macro news, and is any lead large enough to trade after costs?

9.4 million	**75 days**	**453**	**31 tests**
one-minute bars, May 2016 – Oct 2026, four instruments	of one-second data, Jul – Oct 2026 (73 for the ETFs)	macro releases: CPI, payrolls, ISM and FOMC decisions	10 scripts, one command reproduces every table and chart

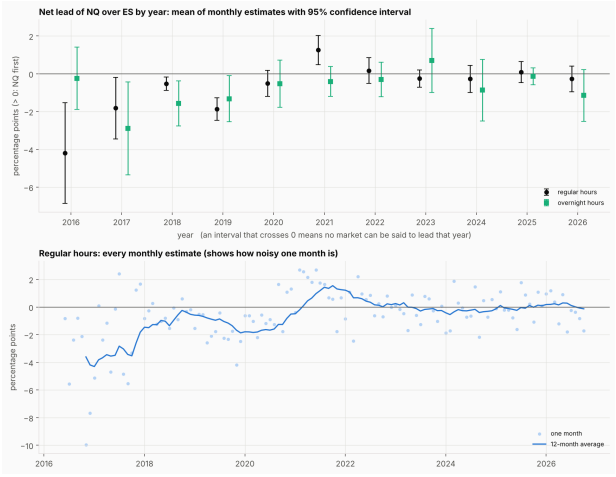
FOUR FINDINGS

1. **The S&P 500 used to lead; today neither index does.**
   In 2016–2019 ES futures led NQ by 1.9 percentage points of the reaction (t = 5.1). Since 2022 the estimate is −0.10 pp [−0.38, 0.17]. The fade tracks liquidity: NQ's share of the two markets' dollar volume rose from about 10% to 35% (t = 4.8).

2. **Futures lead their ETFs, clearly only at one second.**
   At one-minute bars the lead has shrunk to a fraction of a point since 2022. At one-second bars the futures move first on every day of the sample, and NQ leads ES on 91% of days.

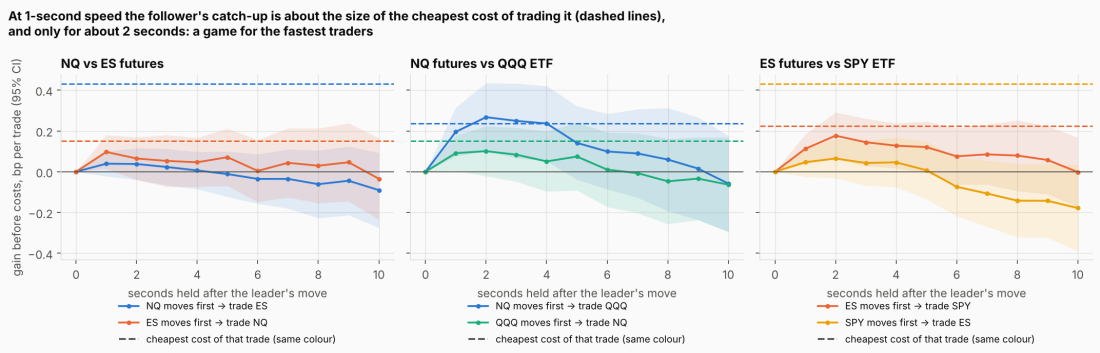
3. **Scheduled news reaches all four markets in the same minute.**
   A CPI minute is 9.5 times as volatile as a normal one (payrolls 6.3×, FOMC 5.7×), and prices follow the surprise: −32 bp per standard deviation of CPI surprise (t = −4.1). No market leads around the releases.

4. **The lead survives exactly up to the cost of trading it.**
   At one-minute speed the follower's catch-up is at most 55% of the cheapest round-trip cost. At one-second speed it roughly equals the cost (0.27 bp against 0.24 bp for QQQ) and is gone after about two seconds, as Grossman and Stiglitz (1980) predict.



Finding 1. Net lead of NQ over ES by year with 95% confidence intervals (top) and every monthly estimate (bottom). Below zero means ES moved first; the lead fades from 2020 onwards.

Validate before you estimate. I first planted known leads in real NQ returns. A textbook Granger test found a lead in 11 of 11 years when there was none, only a coarser price grid; the *net lead* used here recovered a planted 3 pp lead at 3.05–3.15 pp. At one second, a placebo exposed a "staleness trap" (a fake 7.4 pp lead) that comparing only seconds in which both markets trade removes.



Finding 4. Follower's gain before costs (95% CI) by seconds held after the leader's top-0.1% one-second moves, against the cheapest cost of that trade (dashed). The gain only reaches the cost for a couple of seconds: a game for the fastest trading firms, not for a human trader.

WHY IT MATTERS

Market makers need to know which price to watch, hedgers whether a quote is stale, economists whether markets are efficient. Answer: the US index complex is efficient up to about two seconds and one trading cost.

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The full 12-page paper, code, results and tests are on GitHub (release v1.0). Data: London Strategic Edge. Python, with AI coding assistance; the research design and interpretation are the author's.

For research and educational purposes only; not investment advice. 1 bp = 0.01%. pp = percentage points of the follower's total reaction.